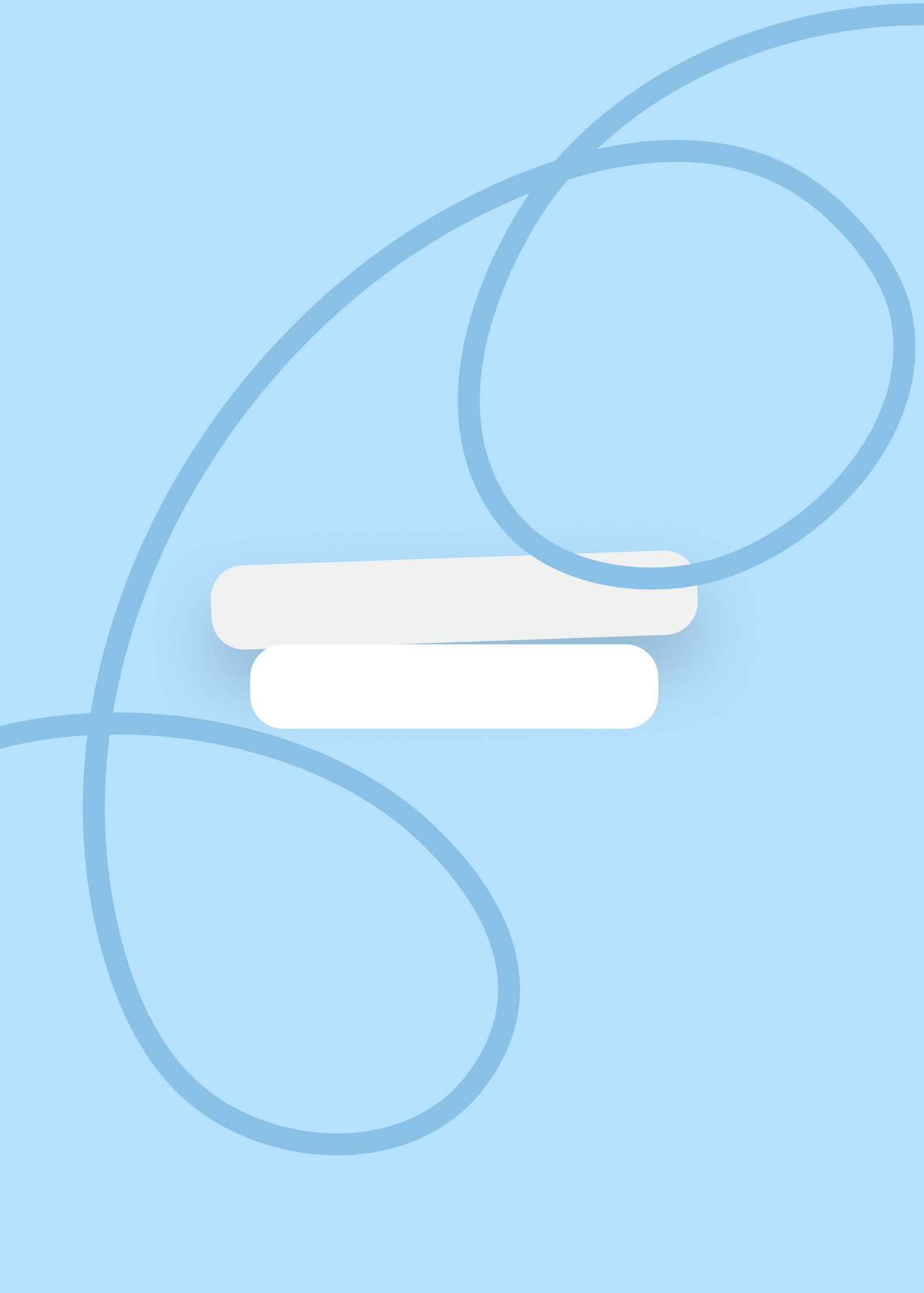


Bilet 1

1. $p \to (q \vee r) \models (p \to q) \vee (q \to r)$

Formula V este o consecință logică a formulei U ($U \models V$) $\Leftrightarrow \forall i : Fp \to \{T,F\}$ a.î. $i(U)=T$ are loc $i(V)=T$

Tabela de adevăr

	p	q	r	q∨r	p→q	q→r	p→(q∨r)	(p→q)∨(q→r)
i_1	T	T	T	T	T	T	T	T
i_2	T	T	F	T	T	F	T	T
i_3	T	F	T	T	F	T	T	T
i_4	T	F	F	F	F	T	F	T
i_5	F	T	T	T	T	T	T	T
i_6	F	T	F	T	T	F	T	T
i_7	F	F	T	T	T	T	T	T
i_8	F	F	F	F	T	T	T	T

$\Rightarrow$ pt $\forall i : \{p,q,r\} \to \{T,F\}$ a.î. $i(U)=T$ are loc $i(V)=T$

Tabelă sem. (arbore binar)

T.c.c : O formulă U este tautologie $\Leftrightarrow \exists$ o tabelă sem închisă pt formula $\neg U$

$p \to (q \vee r) \models (p \to q) \vee (p \to r)$

$(p \to (q \vee r)) \wedge \neg((p \to q) \vee (p \to r))$ (1) ✓

| α pt (1)

$p \to (q \vee r)$ (2) ✓

$\neg((p \to q) \vee (p \to r))$ (3) ✓

| α pt (3)

$\neg(p \to q)$ (4) ✓

$\neg(p \to r)$ (5) ✓

| α pt (4)

p

$\neg q$

| α pt (5)

p

$\neg r$

β pt (2)

$\neg p$ ⊗

$q \vee r$ (6)

β pt (6)

q ⊗ $\quad$ r ⊗

$\neg U$ e închisă $\xrightarrow{T.c.c.}$ U tautologie

2. • $\exists$ fală de v

$(\exists x)(A(x) \vee B(x)) = (\exists x)A(x) \vee (\exists x)B(x)$

• ∀ față de ∧

$(\forall x)(A(x) \wedge B(x)) = (\forall x)A(x) \wedge (\forall x)B(x)$

$\neg((\exists x)(A(x) \vee B(x))$

PAS1 ; Elim $\to$ $\leftrightarrow$

PAS2 DM:

$\neg(\exists x)\neg(A(x) \vee B(x))$

$\equiv (\forall x)\neg(A(x) \vee B(x))$

$\equiv (\forall x)(\neg A(x) \wedge \neg B(x))$

PAS3: Distrib cuantif

$\equiv (\forall x)\neg A(x) \wedge (\forall x)\neg B(x)$

$\equiv \neg((\exists x)A(x)) \wedge \neg((\exists x)B(x))$

$\equiv \neg((\exists x)A(x) \vee (\exists x)B(x))$

Știm că $A(x) \to (A(x) \vee B(x))$ este o taut.

Reg contrapoziției : $\neg(A(x) \vee B(x)) \to \neg A(x)$

Aplicăm reg generalizării. $(\forall x)(\neg(A(x) \vee B(x)) \to \neg A(x))$

$(\forall x)\neg(A(x) \vee B(x)) \to (\forall x)\neg A(x)$

$\neg(\forall x)\neg A(x) \to \neg(\forall x)\neg(A(x) \vee B(x))$

$(\exists x)A(x) \to (\exists x)(A(x) \vee B(x))$ și $((\exists x)A(x) \vee (\exists x)B(x)) \to (\exists x)(A(x) \vee B(x))$

Analog $(\exists x)B(x) \to (\exists x)(A(x) \vee B(x))$

T.C.C. : Fie S o mult. de formule predicative, iar A o form predic.

completitudine: dacă $S \models A$ atunci $S \vdash A$

corectitudine: dacă $S \vdash A$ atunci $S \models A$

3.

x	y	z	$\bar{x}$	$\bar{y}$	$\bar{z}$	$y \oplus z$	$(\bar{x} \uparrow z)$	$(x \downarrow \bar{y})$	$x(y \oplus z)$	$\bar{y}(\bar{x} \uparrow z)$	$(x \downarrow \bar{y})\bar{z}$	$f(x,y,z)$	
0	0	0	1	1	1	0	1	0	0	1	0	1	m_0
0	0	1	1	1	0	1	0	0	0	0	0	0	
0	1	0	1	0	1	1	1	1	0	0	1	1	m_2
0	1	1	1	0	0	0	0	1	0	0	0	0	
1	0	0	0	1	1	0	1	0	0	1	0	1	m_4
1	0	1	0	1	0	1	1	0	1	1	0	1	m_5
1	1	0	0	0	1	1	1	0	1	0	0	1	m_6
1	1	1	0	0	0	0	1	0	0	0	0	0	

$p \uparrow q = \neg(p \wedge q)$

$0 \uparrow 0 = \neg(0 \wedge 0) = \neg 0 = 1$

$f(x,y,z) = x(y \oplus z) \vee \bar{y}(\bar{x} \uparrow z) \vee (x \downarrow \bar{y})\bar{z}$

$\bar{x} = \neg x$

$x \oplus y = \neg(x \leftrightarrow y)$

FCD, $f = 1$

FCC, $f = 0$

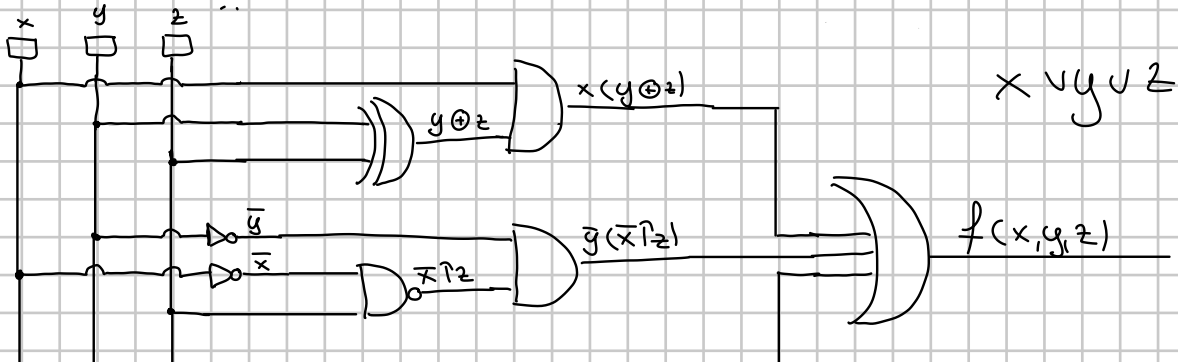
FCD

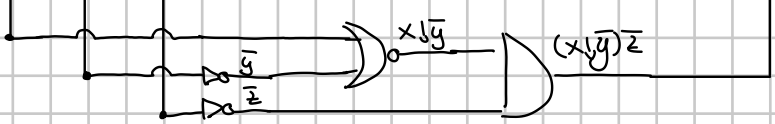
mintermii : $xyz, x\bar{y}z, x\bar{y}\bar{z}, \bar{x}y\bar{z}, \bar{x}\bar{y}z, \bar{x}\bar{y}z$ FCD : $xyz \vee x\bar{y}z$ — —

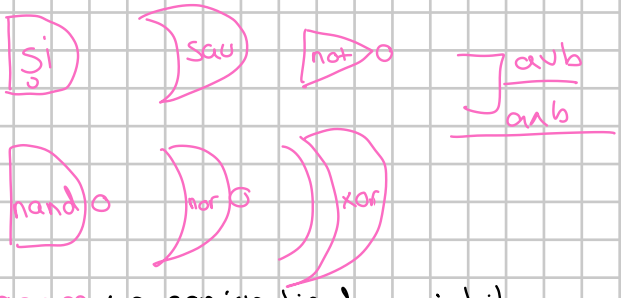
FCC

maxtermii: $xy\bar{z}, x\bar{y}\bar{z}, \bar{x}\bar{y}\bar{z}$ FCC $\bar{x}\bar{y}\bar{z} \wedge$ — —

$f(x,y,z) = x(y \oplus z) \vee \bar{y}(\bar{x} \uparrow z) \vee (x \downarrow \bar{y})\bar{z}$







monom : o conjuncție de variabile
minterm : o conj care conține toate var.
maxterm : o disjuncție care conține toate var.

FCC : conjuncția maxtermilor corespunzători argumentelor pt care funcția val. 0
FCD : disjuncția mintermilor ——— " ——— 1

4 La toți copiii le plac bomboanele

1. $(\forall x)(\forall y)(C(x) \wedge B(y)) \rightarrow P(x,y)$
2. $(\forall x)(\exists y)(B(y) \wedge P(x,y)) \rightarrow \neg N(x)$
3. $(\forall x)(\exists y)(D(y) \wedge M(x,y)) \rightarrow N(x)$
4. $(\forall x)(\forall y)(C(x,y) \wedge D(y) \rightarrow (S(x,y) \vee M(x,y))$
5. $(\exists y)(D(y) \wedge C(ion,y))$
6. $B(lifesavers)$

Concluzie: $C(ion) \rightarrow \exists(y)(D(y) \wedge S(ion,y))$

Pp cl. adv.
ip 1: $C(i) \wedge B(lifes) \rightarrow P(ion, lifes)$
ip 2: $C(i) \wedge B(lifes) \xrightarrow{MP} P(ion, lifes)$
ip 2: $((\exists y)(B(y) \wedge P(ion,y))) \rightarrow N(ion)$
ip 3: $\neg N(ion) \Rightarrow \neg((\exists y)(D(y) \wedge M(ion,y))) \rightarrow (\forall y)(D(y) \rightarrow \neg M(ion,y))$

$S = \{$
$\neg C(x) \vee \neg B(y) \vee P(x,y)$ C_1 ✓
$\neg B(y) \vee \neg P(x,y) \vee \neg F(x)$ C_2 ✓
$\neg D(y) \vee \neg M(x,y) \vee F(x)$ C_3 ✓
$\neg D(y) \vee \neg B(y) \vee S(x,y) \vee M(x,y)$ C_4 ✓
$D(y_1)$ C_5 ✓
$B(a,y)$ C_6 ✓
$B(b)$ C_7 ✓
$C(a)$ C_8
$\neg D(y) \vee \neg S(a,y)$ C_9 ✓

$C_8 = C(a)$ $\quad C_1 = \neg C(x) \vee \neg B(y) \vee P(x,y)$ $[x \leftarrow a]$

$C_{10} = \neg B(y) \vee P(a,y)$ $\quad C_7 = B(b)$ $[y \leftarrow b]$

$C_{11} = P(a,b)$ $\quad C_2 = \neg B(y) \vee \neg P(x,y) \vee \neg F(x)$ $[x \leftarrow a]$ $[y \leftarrow b]$

$C_{12} = \neg B(b) \vee \neg F(a)$ $\quad C_4 = B(b)$

$C_{13} = \neg F(a)$ $\quad C_3 = \neg D(y) \vee \neg M(x,y) \vee F(x)$ $\quad [x \leftarrow a]$

$C_{14} = \neg D(y) \vee \neg M(a,y)$ $\quad C_5 = D(y)$

$C_{15} = \neg M(a,y)$ $\quad C_4 = \neg D(y) \vee \neg Buy(x,y) \vee S(x,y) \vee M(x,y)$ $\quad [x \leftarrow a]$

$C_{16} = \neg D(y) \vee \neg Buy(a,y) \vee S(a,y)$ $\quad C_5 = D(y)$

$C_{17} = \neg Buy(a,y) \vee S(a,y)$ $\quad C_6 = Buy(a,y)$

$C_{17} = S(a,y)$ $\quad C_9 = \neg D(y) \vee \neg S(a,y)$

$C_{18} = \neg D(y)$ $\quad C_5 = D(y)$

$C_{19} = \square$ $\overset{T.C.S.}{\Longrightarrow}$ Concluzia are loc

Bilet 1

1. $\vdash (U \to (V \to Z)) \to (V \to (U \to Z)) = A$

$\neg (U \to (V \to Z)) \to (V \to (U \to Z))$ (1)

| α pt (1)

$U \to (V \to Z)$ (2) ✓

|

$\neg (V \to (U \to Z))$ (3) ✓

| α pt (3)

[V]

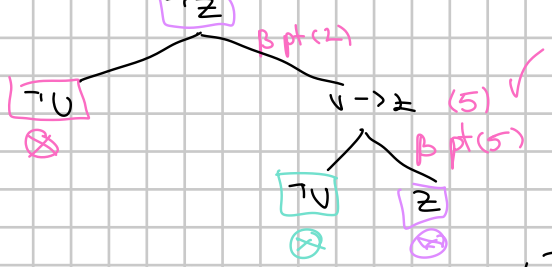
|

$\neg (U \to Z)$ (4) ✓

| α pt (4)

[U]

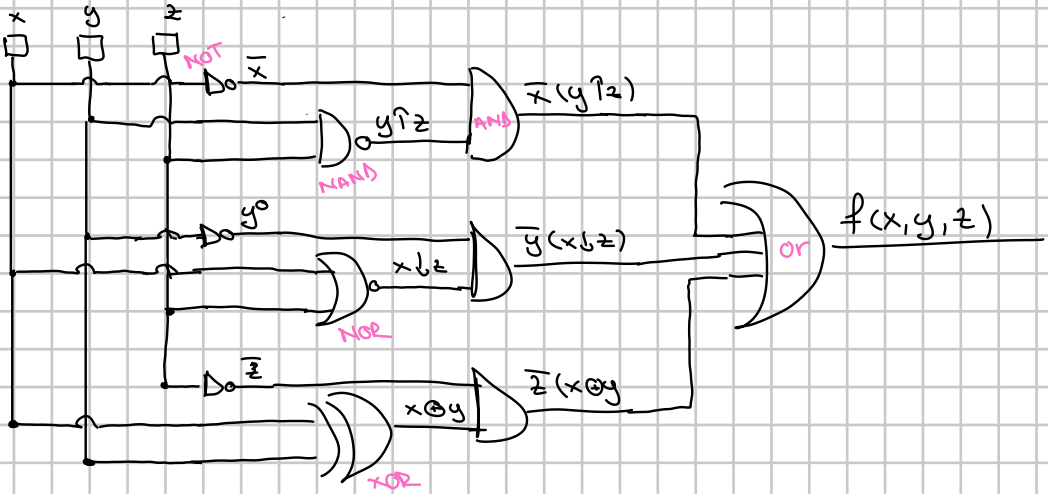
|



toate ram sunt închise $\Rightarrow$ $\neg A$ inconsist $\overset{T.C.S.}{\Rightarrow}$ A tautologie

2 făcut

3. $f(x,y,z) = \bar{x}(y \uparrow z) \vee \bar{y}(x \downarrow z) \vee \bar{z}(x \oplus y)$



x	y	z	$\bar{x}$	$\bar{y}$	$\bar{z}$	$y \uparrow z$	$x \downarrow z$	$x \oplus y$	$\bar{x}(y \uparrow z)$	$\bar{y}(x \downarrow z)$	$\bar{z}(x \oplus y)$	$f(x,y,z)$	
0	0	0	1	1	1	1	1	0	1	1	0	1	m_0
0	0	1	1	1	0	1	0	0	1	0	0	1	m_1
0	1	0	1	0	1	1	1	1	1	0	1	1	m_2
0	1	1	1	0	0	0	0	1	0	0	0	0	
1	0	0	0	1	1	1	0	1	0	0	1	1	m_4
1	0	1	0	1	0	1	0	1	0	0	0	0	
1	1	0	0	0	1	1	0	0	0	0	0	0	
1	1	1	0	0	0	0	0	0	0	0	0	0	

Simplif bi Quine Mc'Clusky

$Sf = \{\overset{m_0}{(0,0,0)}, \overset{m_1}{(0,0,1)}, \overset{m_2}{(0,1,0)}, \overset{m_4}{(1,0,0)}\}$

	x	y	z	
I	0	0	0	m_0 ✓
II	0	0	1	m_1 ✓
	0	1	0	m_2 ✓
	1	0	0	m_4 ✓
Fact simplă III = I + II	0	0	-	$m_0 \vee m_1 = max_1 = \bar{x}\bar{y}$
	0	-	0	$m_0 \vee m_2 = max_2 = \bar{x}\bar{z}$
	-	0	0	$m_0 \vee m_4 = max_3 = \bar{y}\bar{z}$

min \ max	max_1	max_2	max_3
m_0	*	*	*
m_1	(*)		
m_2		(*)	
m_4			(*)

$M(f) = 3 = C(f) \Rightarrow$ Caz I

$f^D(x,y,z) = \bar{x}\bar{y} \vee \bar{x}\bar{z} \vee \bar{y}\bar{z}$

Karnaugh

$Sf = \{\overset{m_0}{(0,0,0)}, \overset{m_1}{(0,0,1)}, \overset{m_2}{(0,1,0)}, \overset{m_4}{(1,0,0)}\}$

x y z

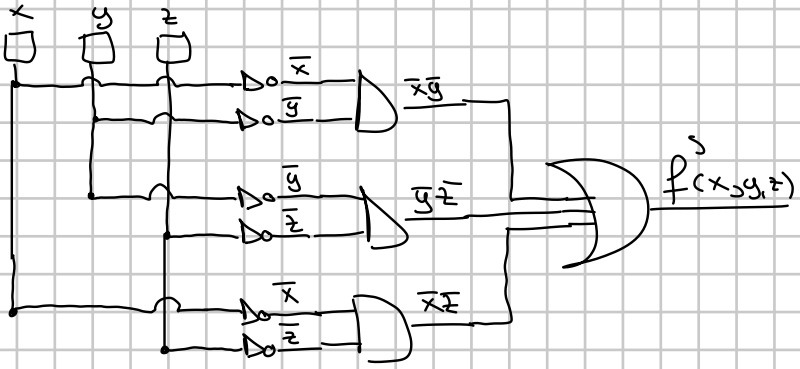
x \ yz	00	01	11	10
0	m_0	m_1		m_2
1	m_4			

$max_1 = m_0 \vee m_1 = \bar{x}\bar{y}$

$max_2 = m_0 \vee m_4 = \bar{y}\bar{z}$

$max_3 = m_0 \vee m_2 = \bar{x}\bar{z}$

$M(f) = 3$, $C(f) = 3$ $\Rightarrow$ Caz I $\quad f^S(x,y,z) = \bar{x}\bar{y} \vee \bar{y}\bar{z} \vee \bar{x}\bar{z}$



Bilet 2

1. $p \to q, \neg(q \to r) \to \neg p \vdash p \to r$

$p \to q \equiv \neg p \vee q$

$\neg(q \to r) \to \neg p \equiv (q \to r) \vee \neg p \equiv \neg q \vee r \vee \neg p$

$p \to r \equiv \neg p \vee r$

$\neg(p \to r) \equiv p \wedge \neg r$

$S = \{ \underset{(1)}{\overset{C_1}{\neg p \vee q}}, \underset{(3)}{\overset{C_2}{\neg q \vee r \vee \neg p}}, \underset{(2)}{\overset{C_3}{\neg p \vee r}}, \underset{(4)}{\overset{C_4}{p}}, \underset{(5)}{\overset{C_5}{\neg r}} \}$

$Res_q^{lock}(C_1, C_2) = \underset{(2)}{\neg p} \vee \underset{(4)}{r} = C_6$

$Res_p^{lock}(C_6, C_4) = \underset{(4)}{r} = C_7$

$Res_r^{lock}(C_7, C_5) = \square \overset{TCC}{\Rightarrow} S$ inconsist.

2. $(\forall x)(A(x) \wedge B(x)) \vdash (\forall x)A(x) \vee (\forall x)B(x)$

$(\forall x)(A(x) \wedge B(x)) \wedge (\exists x)\neg A(x) \wedge (\exists x)\neg B(x)$ (1)

| α pt (1)

$(\forall x)Ax \wedge (\forall x)B(x)$ (2) ✓

|

$(\exists x)\neg A(x) \wedge (\exists x)\neg B(x)$ (3)

| α pt (2)

$(\forall x)A(x)$ (4)

|

$(\forall x)B(x)$ (5)

| α pt (3)

$(\exists x)\neg A(x)$ (6) ✓

|

$(\exists x)\neg B(x)$ (7) ✓

| δ pt (6) ct a

$\neg A(a)$

| δ pt ($\exists$), ct b

$\neg B(b)$

| $\forall$ pt (4) ct a.

$A(a)$

A(6)

(∀ x A(x)

3. $f(x_1,x_2,x_3) = x_1x_3 \vee x_1x_2x_3 \vee \overline{x_1}x_2x_3$

Bilet sup.

1. $U = p \vee \neg(q \wedge \neg r \to p) \to p \wedge q \wedge \neg r$

p	q	r	¬r	q∧¬r	q∧¬r→p	¬(q∧¬r→p)	p∨¬(q∧¬r→p)	p∧q	p∧q∧¬r	U	
T	T	T	F	F	T	F	T	T	F	F	i_1
T	T	F	T	T	T	F	T	T	T	T	i_2
T	F	T	F	F	T	F	T	F	F	F	i_3
T	F	F	T	F	T	F	T	F	F	F	i_4
F	T	T	F	F	T	F	F	F	F	T	i_5
F	T	F	T	T	F	T	T	F	F	F	i_6
F	F	T	F	F	T	F	F	F	F	T	i_7
F	F	F	T	F	T	F	F	F	F	T	i_8

modele: i_2, i_5, i_4, i_8 anti-modele: i_1, i_3, i_4, i_6

3. $f(x_1,x_2,x_3,x_4) = \underset{0\,0\,0\,1}{m_1}\overline{x_1}\overline{x_2}\overline{x_3}x_4 \vee \underset{m_{15}}{}x_1x_2x_3x_4 \vee \underset{1\,0\,1\,0}{m_{10}}x_1\overline{x_2}x_3\overline{x_4} \vee \underset{1\,1\,0\,1}{m_{13}}x_1x_2\overline{x_3}x_4 \vee \underset{0\,0\,1\,1}{m_3}\overline{x_1}\overline{x_2}x_3x_4 \vee \underset{1\,1\,0\,0}{m_{12}}x_1x_2\overline{x_3}\overline{x_4} \vee \underset{m_0}{}\overline{x_1}\overline{x_2}\overline{x_3}\overline{x_4} \vee \underset{1\,0\,1\,1}{m_{11}}x_1\overline{x_2}x_3x_4 \vee \underset{m_2}{\underset{m_{14}}{}}x_1x_2x_3\overline{x_4} \vee \overline{x_1}\overline{x_2}x_3\overline{x_4}$

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01				
11	m_{12}	m_{13}	m_{15}	m_{14}
10			m_{11}	m_{10}

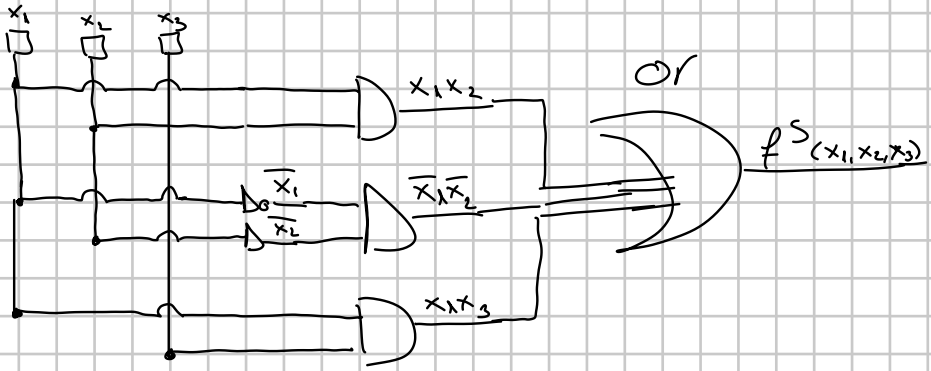
$\max_1 = m_{12} \vee m_{13} \vee m_{14} \vee m_{15} = x_1x_2$

$\max_2 = m_0 \vee m_1 \vee m_3 \vee m_2 = \overline{x_1}\overline{x_2}$

$\max_3 = m_{15} \vee m_{14} \vee m_{11} \vee m_{10} = x_1x_3$

$M(f) = \{\max_1, \max_2, \max_3\}$ $C(f) = \{\max_1, \max_2, \max_3\}$ $\Big\}$ $M(f) = C(f) \to$ caz al I

$f^S(x_1,x_2,x_3) = x_1x_2 \vee \overline{x_1}\overline{x_2} \vee x_1x_3$



Bilet 4 sup.

1. $p \wedge (q \to r) \equiv (p \wedge q) \to (p \wedge r)$

p	q	r	q→r	p∧q	p∧r	p∧(q→r)	(p∧q)→(p∧r)
T	T	T	T	T	T	T	T
T	T	F	F	T	F	F	F

T	F	T	T	F	T	T	T
T	F	F	T	F	F	T	T
F	T	T	T	F	F	F	T
F	T	F	F	F	F	F	T
F	F	T	T	F	F	F	T
F	F	F	T	F	F	F	T

$\Rightarrow$ nu se verif distrib ——"——

2. $A_1 = P(a, h(x,u), g(z))$
$A_2 = P(y, h(y, f(z)), g(x))$

P1: Verif dacă au același pred ✓
P2: Verif dacă au aceeași aritate ✓
P3:

$\theta_1 = [y \leftarrow a]$
Aplicăm
$\theta_1(A_1) = P(a, h(x,u), g(z))$
$\theta_1(A_2) = P(a, h(a, f(z)), g(x))$

$\theta_2 = [x \leftarrow a]$
$\theta_2(\theta_1(A_1)) = P(a, h(a,u), g(z))$
$\theta_2(\theta_1(A_2)) = P(a, h(a, f(z)), g(a))$

$\theta_3 = [u \leftarrow f(z)]$
$\theta_3(\theta_2(\theta_1(A_1))) = P(a, h(a, f(z)), g(z))$
$\theta_3(\theta_2(\theta_1(A_2))) = P(a, h(a, f(z)), g(a))$

$\theta_4 = [z \leftarrow a]$
$\theta_4(\theta_3(\theta_2(\theta_1(A_1)))) = P(a, h(a, f(a)), g(a))$ } Au devidentice $\Rightarrow A_1, A_2$ sunt unificabile
$\theta_4(\theta_3(\theta_2(\theta_1(A_2)))) = P(a, h(a, f(a)), g(a))$

$mgu(A_1, A_2) = \theta_1 \circ \theta_2 \circ \theta_3 \circ \theta_4 = [y \leftarrow a, x \leftarrow a, u \leftarrow f(z), z \leftarrow a]$

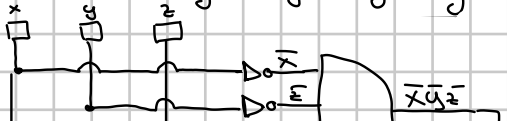
$A_3 = P(x, f(a), b)$
$A_4 = P(y, z, g(b))$
P1 ✓ P2 ✓
P3: $\theta_1 = [x \leftarrow y]$
$\theta_1(A_3) = P(y, f(a), b)$
$\theta_1(A_4) = P(y, z, g(b))$

$\theta_2 = [z \leftarrow f(a)]$
$\theta_2(\theta_1(A_3)) = P(y, f(a), b)$ } $\rightarrow$ Nu se poate substitui b cu $f(b)$ $\rightarrow A_3$ și A_4 nu sunt unificabili
$\theta_2(\theta_1(A_4)) = P(y, f(a), g(b))$

$A_5 = P(x, g(x), f(x))$
$A_6 = P(y, z, y)$
P1 ✓ P2 ✓ aritate
P3: $\theta_1 = [x \leftarrow y]$
$\theta_1(A_5) = P(y, g(y), f(y))$
$\theta_1(A_6) = P(y, z, y)$

$\theta_2 = [z \leftarrow g(y)]$
$\theta_2(\theta_1(A_5)) = P(y, g(y), f(y))$ } $\Rightarrow$ —— ——— A_5, A_6 nu sunt unificabile
$\theta_2(\theta_1(A_6)) = P(y, g(y), y)$

3. $f(x,y,z) = m_0 \vee m_1 \vee m_2 \vee m_3 \vee m_5 \vee m_6$
$= \bar{x}\bar{y}\bar{z} \vee \bar{x}\bar{y}z \vee \bar{x}y\bar{z} \vee \bar{x}yz \vee x\bar{y}z \vee xy\bar{z}$





$Sf = \{(0,0,0)^{m_0}, (0,0,1)^{m_1}, (0,1,0)^{m_2}, (0,1,1)^{m_3}, (1,0,1)^{m_5}, (1,1,0)^{m_6}\}$

	x	y	z	
I	0	0	0	m_0 ✓
II	0	0	1	m_1 ✓
	0	1	0	m_2 ✓
III	0	1	1	m_3 ✓
	1	0	1	m_5 ✓
	1	1	0	m_6 ✓
Fact. simple	0	0	–	$m_0 \vee m_1$ ✓
IV = I + II	0	–	0	$m_0 \vee m_2$ ✓
V = II + III	0	–	1	$m_1 \vee m_3$ ✓
	–	0	1	$m_1 \vee m_5 = max_1 = \bar{y}z$
	0	1	–	$m_2 \vee m_3$ ✓
	–	1	0	$m_2 \vee m_6 = max_2 = y\bar{z}$
Fact dublu	0	–	–	$m_0 \vee m_1 \vee m_2 \vee m_3 = m_0 \vee m_1 \vee m_2 \vee m_3 = max_3 = \bar{x}$
VI = IV + V	0	–	–	$m_0 \vee m_1 \vee m_2 \vee m_3 = \bar{m}_0 \vee m_2 \vee m_1 \vee m_3$

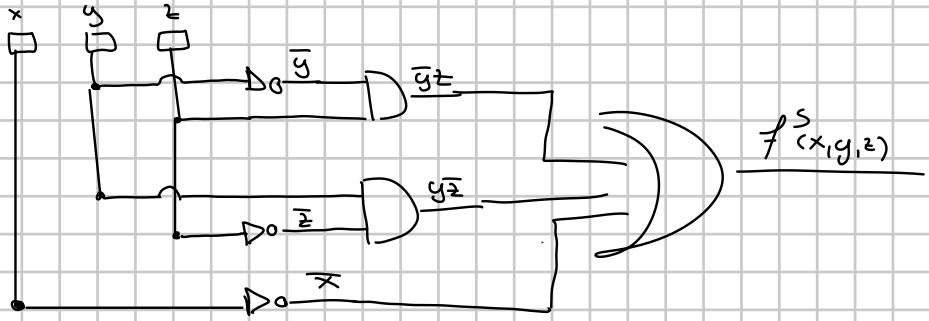
$M(f) = \{max_1, \dots, max_3\}$

mon. / mint.	max_1	max_2	max_3
m_0			(*)
m_1	*		*
m_2		*	*
m_3			(*)
m_5	(*)		
m_6		(*)	

$C(f) = \{max_1, max_2, max_3\}$

$M(f) = C(f) \Rightarrow$ caz I

$f^S(x,y,z) = \bar{y}z \vee y\bar{z} \vee \bar{x}$



Bilet 5

1. $p \wedge q, \neg p \vee \neg q \vee r, r \to t \vdash t$

$p = c_1$
$q = c_2$
$\neg p \vee \neg q \vee r = c_3$
$r \to t = \neg r \vee t = c_4$
$\neg t = c_5$
$Y = \{c_5\}$ multimea suport al lui S
$S = \{p^{c_1}, q^{c_2}, \neg p \vee \neg q \vee r^{c_3}, \neg r \vee t^{c_4}, \neg t^{c_5}\}$
$S \setminus Y = \{c_1, c_2, c_3, c_4\}$ consist

$c_5 - Res_t(c_4, c_5) = \neg r$
$c_6 = Res_r(c_5, c_3) = \neg p \vee \neg q$

$C_7 = Res_g(C_6, C_1) = \neg P$
$C_8 = Res_P(C_7, C_1) = \square$ r.cc. $\Rightarrow$ S inconsist

2. $S = \{\underset{C_1}{\neg P(x) \vee Q(x) \vee \neg R(x)}, \underset{C_2}{P(f(a))}, \underset{C_3}{\neg Q(x)}, \underset{C_4}{\neg W(y)}, \underset{C_5}{R(y) \vee W(y)}\}$

$C_5 = R(y) \vee W(y)$ — $C_4 = \neg W(y)$ unit

unit $C_6 = R(y)$ — $C_1 = \neg P(x) \vee Q(x) \vee \neg R(x)$ $[y \leftarrow x]$

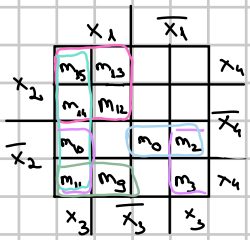
$C_7 = \neg P(x) \vee Q(x)$ — $C_3 = \neg Q(x)$ unit

unit $C_8 = \neg P(x)$ — $C_2 = P(f(a))$ unit $[x \leftarrow f(a)]$

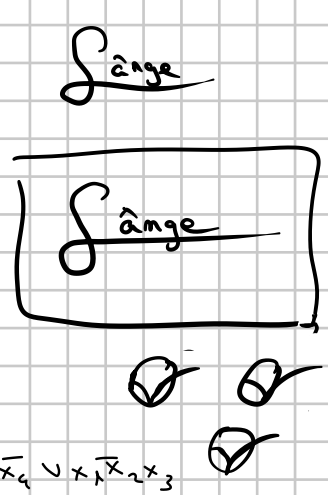
$C_9 = \square$ r.c.c. $\Rightarrow$ S e inconsist DA. ∃

rafinare / rezolutie unit si input

3. $f(x_1, x_2, x_3, x_4) = \underset{m_9}{x_1\bar{x}_2\bar{x}_3x_4} \vee \underset{m_{13}}{x_1x_2\bar{x}_3x_4} \vee \underset{m_{10}}{x_1\bar{x}_2x_3\bar{x}_4} \vee \underset{m_{15}}{x_1x_2x_3x_4} \vee \underset{m_{12}}{x_1x_2\bar{x}_3\bar{x}_4} \vee \underset{m_3}{\bar{x}_1\bar{x}_2x_3x_4} \vee \underset{m_{14}}{x_1x_2x_3\bar{x}_4} \vee \underset{m_0}{\bar{x}_1\bar{x}_2\bar{x}_3\bar{x}_4} \vee \underset{m_{11}}{x_1\bar{x}_2x_3x_4} \vee \underset{m_2}{\bar{x}_1\bar{x}_2x_3\bar{x}_4}$



$max_1 = m_{15} \vee m_{13} \vee m_{14} \vee m_{12} = x_1x_2$
$max_2 = m_{15} \vee m_{14} \vee m_{10} \vee m_{11} = x_1x_3$
$max_3 = m_{10} \vee m_{11} \vee m_2 \vee m_3 = \bar{x}_2x_3$
$max_4 = m_0 \vee m_2 = \bar{x}_1\bar{x}_3\bar{x}_4$
$max_5 = m_9 \vee m_{11} = x_1\bar{x}_2x_3$



$M(f) = \{max_1, \dots, max_5\}$
$C(f) = \{max_1, max_3, max_4, max_5\}$
$M(f) \neq C(f) \Rightarrow$ Cazul II

$g(x_1, x_2, x_3, x_4) = \{max_1, max_3, max_4, max_5\}$
$h(x_1, x_2, x_3, x_4) = max_2$
$f^S(x_1, __) = g(__) \vee R(__) = x_1x_2 \vee x_1x_3 \vee \bar{x}_2x_3 \vee \bar{x}_1\bar{x}_2\bar{x}_4 \vee x_1\bar{x}_2x_3$

Bilet C

1. $U = (p \to (q \to r)) \wedge \neg((p \to q) \to (p \to r))$

Pas 1: Înlocuire $\to$
$U \equiv (\neg p \vee (q \to r)) \wedge \neg(\neg(p \to q) \vee (p \to r))$
$U \equiv (\neg p \vee (\neg q \vee r)) \wedge \neg(\neg(\neg p \vee q) \vee (\neg p \vee r))$

Pas 2: DM
$U \equiv (\neg p \vee (\neg q \vee r)) \wedge (\neg(\neg p \vee q) \wedge \neg(\neg p \vee r))$
$U \equiv (\neg p \vee \neg q \vee r) \wedge (\neg\neg p \wedge \neg q) \wedge p \wedge \neg r$

PAS 3: Distrib $\wedge$ față de $\vee$
$U \equiv (\neg p \wedge \neg\neg p \wedge p \wedge \neg r) \vee (\neg p \wedge q \wedge p \wedge \neg r) \vee (\neg q \wedge \neg\neg p \wedge p \wedge \neg r) \wedge (\neg q \wedge q \wedge p \wedge \neg r) \vee (r \wedge \neg\neg p \wedge p \wedge \neg r) \vee (r \wedge q \wedge p \wedge \neg r)$

FND cu 6 cuburi U inconsist.

$\neg U \equiv (p \vee \neg p \vee \neg p \vee r) \wedge (p \vee \neg q \vee \neg p \vee r) \wedge (q \vee p \vee \neg p \vee r) \wedge (q \vee \neg q \vee \neg p \vee r) \wedge (\neg r \vee p \vee \neg p \vee r) \wedge (\neg r \vee q \vee p \vee r)$

FNC cu 6 clauze

Pop ≡ Nesimțire

2. $S = \{\underset{C_1}{\neg P(x) \vee Q(x)}, \underset{C_2}{P(a)}, \underset{C_3}{\neg Q(x) \vee \neg R(x)}, \underset{C_4}{\neg W(a)}, \underset{C_5}{R(y) \vee W(y)}\}$

$C_6 = Res_p^{lock}(C_1, C_2) = Q(a)$ (2)

$C_7 = Res_q^{lock}(C_6, C_3) = \neg R(a)$ (5)

$C_8 = Res_r^{lock}(C_5, C_7) = W(a)$

$C_9 = Res_w^{lock}(C_4, C_8) = \square \overset{T.C.C.}{\Longrightarrow}$ Inconsistentă

3. $f(1,0,0) = f(0,1,1) = 0$ (m_4: $x_1\bar{x}_2\bar{x}_3$; m_3: $\bar{x}_1x_2x_3$)

$f(x_1,x_2,x_3) = m_0 \vee m_1 \vee m_2 \vee m_5 \vee m_6 \vee m_7$

$= \bar{x}_1\bar{x}_2\bar{x}_3 \vee \bar{x}_1\bar{x}_2x_3 \vee \bar{x}_1x_2\bar{x}_3 \vee x_1\bar{x}_2x_3 \vee x_1x_2\bar{x}_3 \vee x_1x_2x_3$

(m_0, m_1, m_2, m_5, m_6, m_7)



$max_1 = m_7 \vee m_6 = x_1x_2$

$max_2 = m_6 \vee m_2 = x_1\bar{x}_3$

$max_3 = m_5 \vee m_7 = x_1x_3$

$max_4 = m_0 \vee m_1 = \bar{x}_1\bar{x}_2$

$max_5 = m_1 \vee m_5 = \bar{x}_2x_3$

$max_6 = m_0 \vee m_1 = \bar{x}_1\bar{x}_3$

m6 epic save !!!

$M(f) = \{max_1 \text{ —— } max_6\}$

$C(f) = \{max_1, max_2, max_3, max_4, max_5, max_6\}$ } $M(f) = C(f) \Rightarrow$ CoA I

$f^S(x_1,x_2,x_3) = x_1x_2 \vee x_2\bar{x}_3 \vee x_1x_3 \vee \bar{x}_1\bar{x}_2 \vee \bar{x}_2x_3 \vee \bar{x}_1\bar{x}_3$